Database project on weather reports and storing history of climate change throughout the year

Objectives:

1. To collect the average temperature , humidity and precipitation every month according to the seasons.
2. To store the details of gradual weather change in different seasons and compare it.
3. To determine a overview of climate change in a focused region.
4. To apply Pl SQl on the project to make it more user friendly.

Introduction:

This project focuses on developing a comprehensive **Weather Reporting Database System** designed to store, manage, and retrieve weather data based on seasonal variations and geographic locations. The system utilizes **SQL** (Structured Query Language) to perform essential database operations, including data storage, retrieval, modification, and analysis. The goal of this project is to create an efficient and scalable solution for tracking weather conditions across various locations, incorporating data such as **temperature**, **precipitation**, **humidity**, and seasonal patterns like **summer**, **monsoon**, and **winter**.

The database system consists of multiple interrelated tables, each representing different aspects of weather data:

- **Location**: Stores information about geographical areas, including divisions, cities, and their corresponding latitude and longitude.
- **Precipitation**: Records waterbody levels and precipitation data for various cities.
- **Temperature**: Tracks the temperature values for cities across different times and seasons.
- **Humidity**: Maintains humidity levels linked to cities and their respective waterbody levels.
- **Seasonal Data**: Includes specific weather data for **summer**, **monsoon**, and **winter** seasons, capturing temperature and humidity variations for each division.

The project also implements **foreign key constraints** to ensure referential integrity between tables, maintaining the relationships between location data, weather parameters, and seasonal information. With these components, the system enables users to query detailed reports on weather patterns by location and season, facilitating informed decision-making for activities dependent on climate conditions.

Through the use of SQL queries, this project demonstrates how relational database principles can be applied to solve real-world problems in environmental monitoring, enabling the efficient storage and analysis of weather-related data. Few operations were applied on to the project and they are:

Creating table:

```
-- Create Location Table
CREATE TABLE Location (
    division VARCHAR(50) PRIMARY KEY ,
    cityname VARCHAR(50) UNIQUE,
    latitude DECIMAL(9, 6),
    longitude DECIMAL(9, 6)
);
```

Insert into table:

```
-- Insert data into Temperature
INSERT INTO Temperature (temp_id, value_t, cityname) VALUES (1, 34.5, 'Teligati');
INSERT INTO Temperature (temp_id, value_t, cityname) VALUES (2, 37.5, 'CRB');
INSERT INTO Temperature (temp_id, value_t, cityname) VALUES (3, 39.5, 'Mirpur');
INSERT INTO Temperature (temp_id, value_t, cityname) VALUES (4, 32.5, 'T_barriage');
```

Select operation:

```
-- Select all data from Location
SELECT * FROM Location;

-- Select specific columns from Precipitation
SELECT cityname, waterlevel FROM Precipitation WHERE waterlevel > 12;

-- Select average temperature from Summer
SELECT AVG(value_t) AS avg_summer_temp FROM Summer;
```

Delete operation:

```
-- Delete all records from Winter for a specific division
DELETE FROM Winter
WHERE division = 'Khulna';
DELETE FROM Temperature
WHERE cityname = 'Teligati';
```

Altering Table :

```
-- Add a new column to Temperature
ALTER TABLE Temperature
ADD recorded_date DATE;
-- Modify the column data type in Humidity
ALTER TABLE Humidity
MODIFY value_h DECIMAL(6, 3);
```

Group operation:

```
-- Group by division and get the average temperature for each division in Summer
SELECT division, AVG(value_t) AS avg_temp
FROM monsoon
GROUP BY division;

-- Group by city and count the number of precipitation records
SELECT cityname, COUNT(*) AS precipitation_count
FROM Precipitation
GROUP BY cityname;;
```

Union operation:

```
-- UNION: set operation Combine data from Summer and Winter for a division
SELECT month_s AS month, division, value_t, value_h
FROM Summer
WHERE division = 'Khulna'
UNION
SELECT month_w AS month, division, value_t, value_h
FROM Winter
WHERE division = 'Khulna';
```

Join operation:

```
-- Join Location and Precipitation on cityname
SELECT l.cityname, l.latitude, l.longitude, p.waterlevel
FROM Location l
INNER JOIN Precipitation p ON l.cityname = p.cityname;

-- Left join Temperature with Location
SELECT l.cityname, t.value_t
FROM Location l
LEFT JOIN Temperature t ON l.cityname = t.cityname;

-- Cross join Summer and Winter
SELECT s.month_s, w.month_w, s.division, w.value_h
FROM Summer s
CROSS JOIN Winter w;
```

PL SQL Operation:

```
--Avg of summer , winter, and monsoon


DBMS_OUTPUT.PUT_LINE('Starting Seasonal Averages Calculation');
DBMS_OUTPUT.PUT_LINE('-----------------------------------');

DECLARE
  -- Variables for storing averages (use %TYPE for dynamic data type)
  avg_temp Summer.value_t%TYPE;
  avg_humidity Summer.value_h%TYPE;

BEGIN
```

```
  DBMS_OUTPUT.PUT_LINE('Starting Seasonal Averages Calculation');
  DBMS_OUTPUT.PUT_LINE('-----------------------------------');

  -- Calculate averages for Summer
  SELECT AVG(value_t), AVG(value_h)
  INTO avg_temp, avg_humidity
  FROM Summer;

  DBMS_OUTPUT.PUT_LINE('Season: Summer');
  DBMS_OUTPUT.PUT_LINE('Average Temperature: ' || avg_temp);
  DBMS_OUTPUT.PUT_LINE('Average Humidity: ' || avg_humidity);

  -- Calculate averages for Monsoon
  SELECT AVG(value_t), AVG(value_h)
  INTO avg_temp, avg_humidity
  FROM Monsoon;

  DBMS_OUTPUT.PUT_LINE('Season: Monsoon');
  DBMS_OUTPUT.PUT_LINE('Average Temperature: ' || avg_temp);
  DBMS_OUTPUT.PUT_LINE('Average Humidity: ' || avg_humidity);

  -- Calculate averages for Winter
  SELECT AVG(value_t), AVG(value_h)
  INTO avg_temp, avg_humidity
  FROM Winter;

  DBMS_OUTPUT.PUT_LINE('Season: Winter');
  DBMS_OUTPUT.PUT_LINE('Average Temperature: ' || avg_temp);
  DBMS_OUTPUT.PUT_LINE('Average Humidity: ' || avg_humidity);
END;
/
```

Updating

```
--update

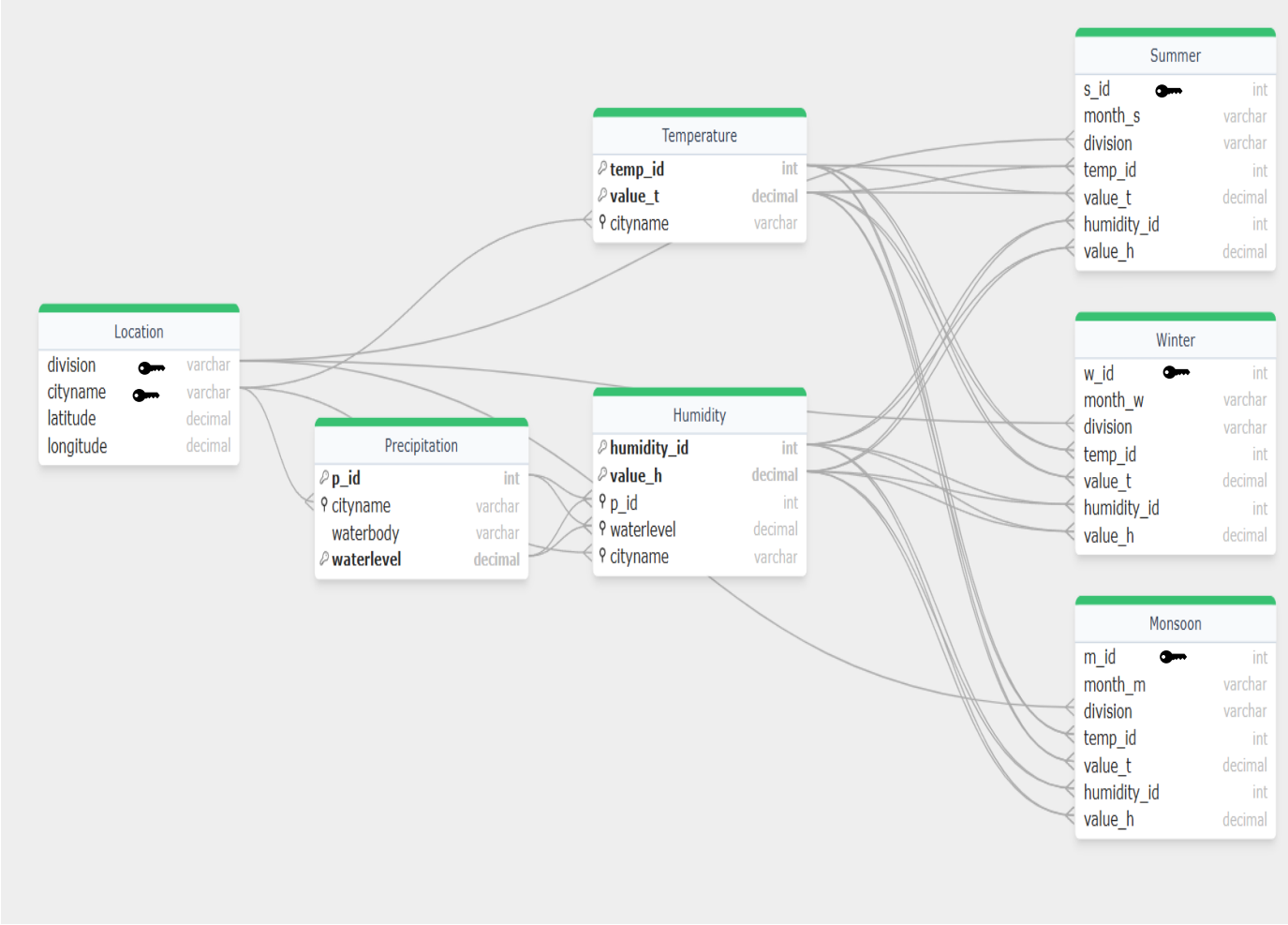
SET SERVEROUTPUT ON;

DECLARE
   -- Variable to hold the new month name
   new_month VARCHAR(20) := 'July';  -- Example: update the month to 'July'
BEGIN
   -- Update the month_s for s_id = 10
   UPDATE Summer
   SET month_s = new_month
   WHERE s_id = 10;
```

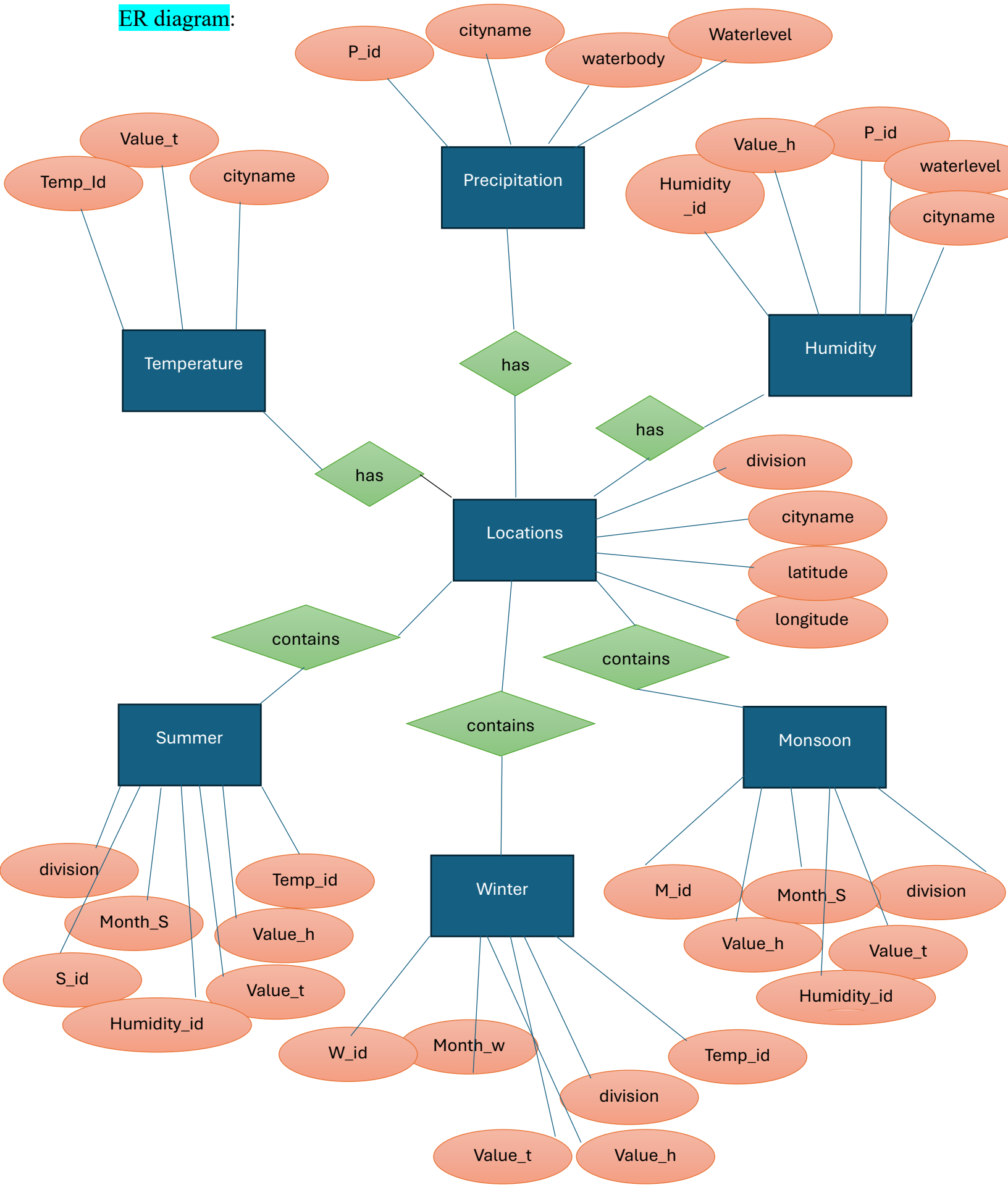
```
    -- Output the result
    DBMS_OUTPUT.PUT_LINE('Month name updated to ' || new_month || ' for s_id = 10.');

    -- Commit the changes
    COMMIT;
END;
/
```

Schema diagram:



ER diagram:


P_id
cityname
waterbody
Waterlevel
Precipitation
Value_t
Temp_Id
cityname
Temperature
Value_h
P_id
Humidity _id
waterlevel
cityname
Humidity
has
has
has
Locations
division
cityname
latitude
longitude
contains
contains
contains
Summer
Winter
Monsoon
division
Temp_id
Month_S
Value_h
S_id
Value_t
Humidity_id
W_id
Month_w
Temp_id
division
Value_t
Value_h
M_id
Month_S
division
Value_h
Value_t
Humidity_id

Conclusion:

From this project we have accomplished the management of data collection of weather reports and climate change. Weather reporting and historical data collection play a crucial role in understanding climate patterns, predicting future weather conditions, and aiding decision-making in agriculture, disaster management, and urban planning. By systematically organizing weather data—such as temperature, humidity, and precipitation—across various seasons and locations, we gain valuable insights into environmental trends and anomalies. his not only supports research and forecasting but also contributes to developing resilient strategies against climate change. Historical weather records serve as a foundation for predictive models, enhancing our ability to mitigate the impact of extreme weather events.

Reference: 1. https://sqlflow.gudusoft.com/#/



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